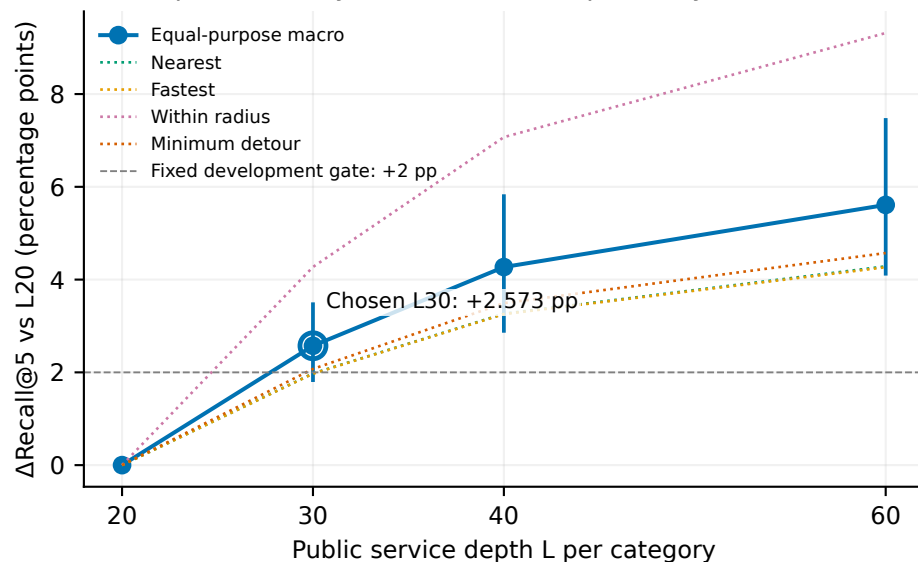


Geo-I: fixed Q, public response-depth utility / cost — fresh confirmation pending

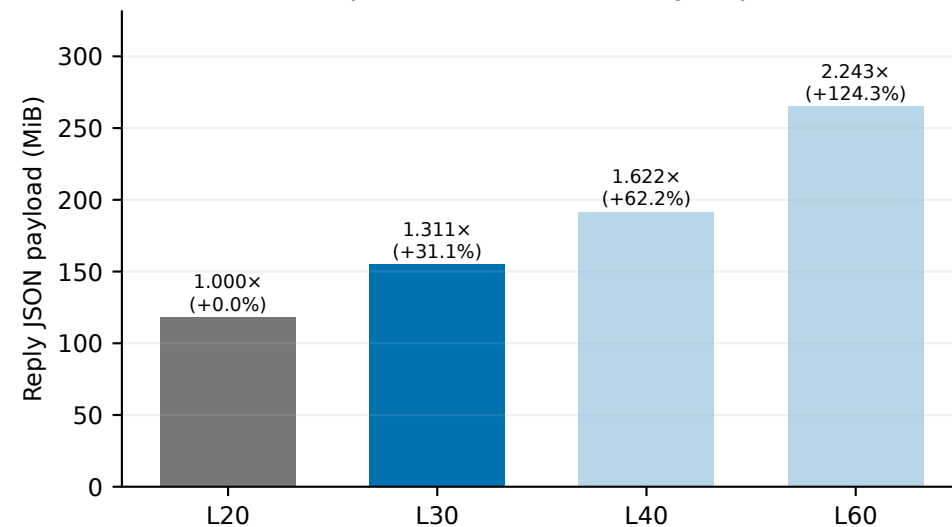
A DEVELOPMENT utility: 6 families

Saved paired-family95% macro CI; exploratory



B DEVELOPMENT response cost

Same Q/clock/request count; additional bytes paid



C Purpose utility and reference coverage: empty reference stays N/A

Purpose	DEV L20 Recall	DEV L30 Recall	DEV reference coverage
Nearest	94.14%	96.12%	27,216/27,216 (100.0%) windows 4,536/4,536
Fastest	94.16%	96.13%	27,216/27,216 (100.0%) windows 4,536/4,536
Within radius	86.60%	90.87%	10,008/27,216 (36.8%) windows 4,506/4,536
Minimum detour	93.85%	95.92%	27,216/27,216 (100.0%) windows 4,536/4,536

Current-only conditional static-service utility. L is per category, for each of 5 Q; application JSON excludes HTTP/TLS.

Development tuning and intervals are exploratory. Paid response depth is not a novel Q algorithm, a privacy improvement or a same-cost SOTA comparison.